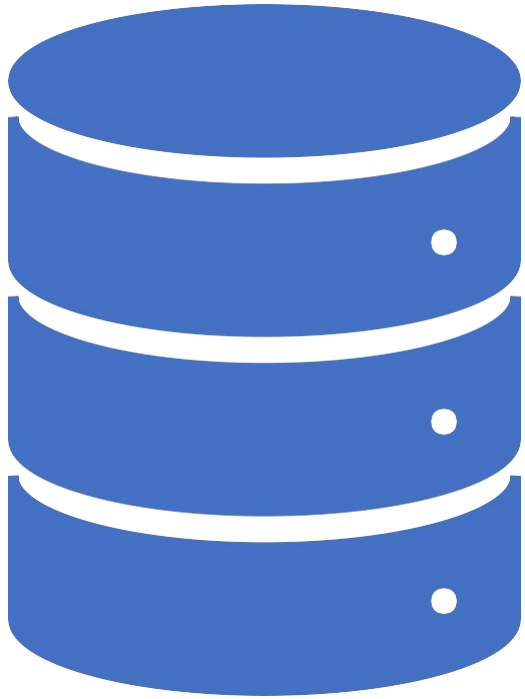


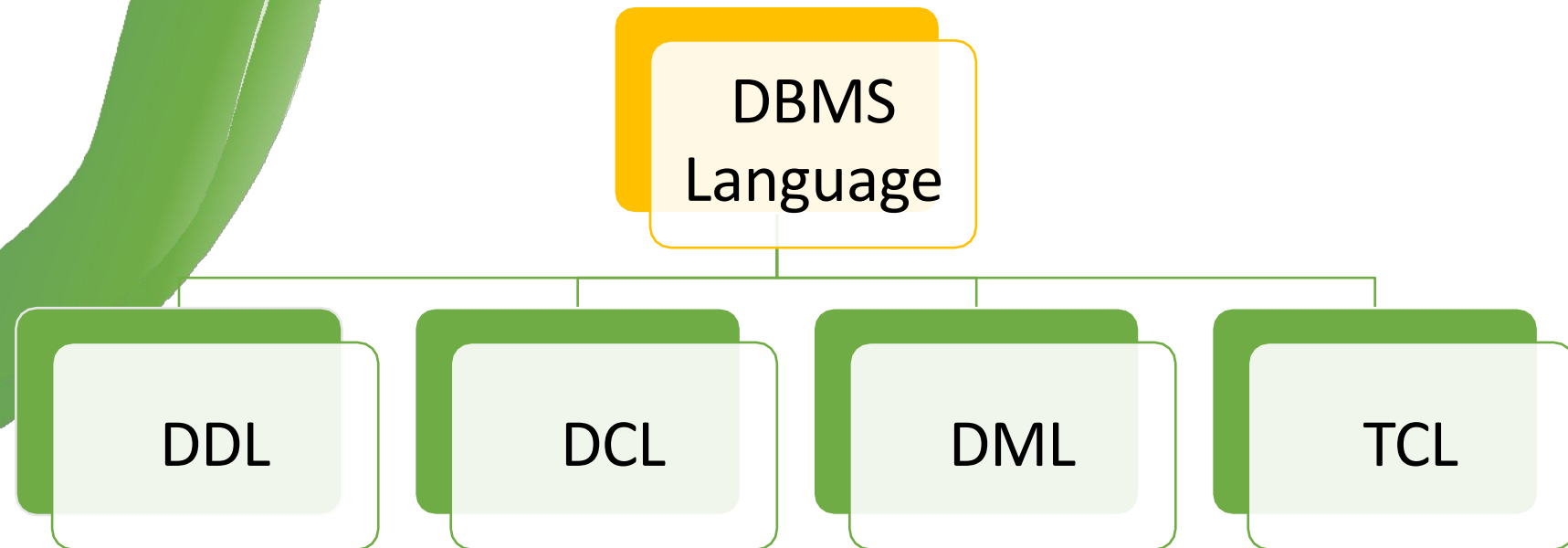


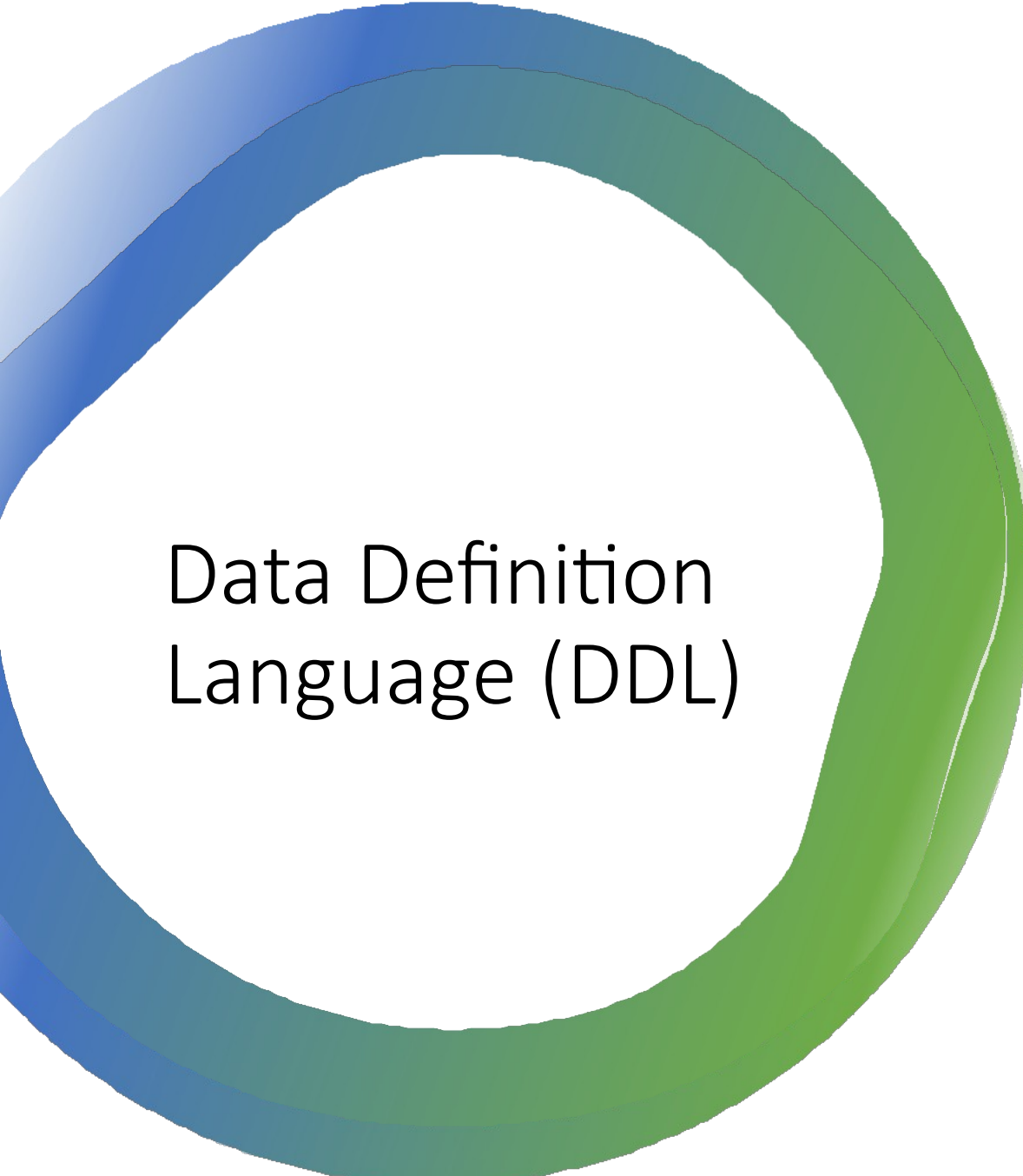
Database Systems

[DB Languages in DBMS]

Database Languages in DBMS

- A DBMS has appropriate languages and interfaces to express database queries and updates.
- Database languages can be used to read, store and update the data in the database.





Data Definition Language (DDL)

- **DDL** stands for **Data Definition Language**.
- It is used to define database structure or pattern.
- It is used to create schema, tables, indexes, constraints, etc. in the database.
- Using the DDL statements, you can create the skeleton of the database.
- DDL is used to store the information of metadata like the number of tables and schemas, their names, indexes, columns in each table, constraints, etc.



Tasks come under DDL

Create:

- It is used to create objects in the database.

Alter:

- It is used to alter the structure of the database.

Drop:

- It is used to delete objects from the database.

Truncate:

- It is used to remove all records from a table.

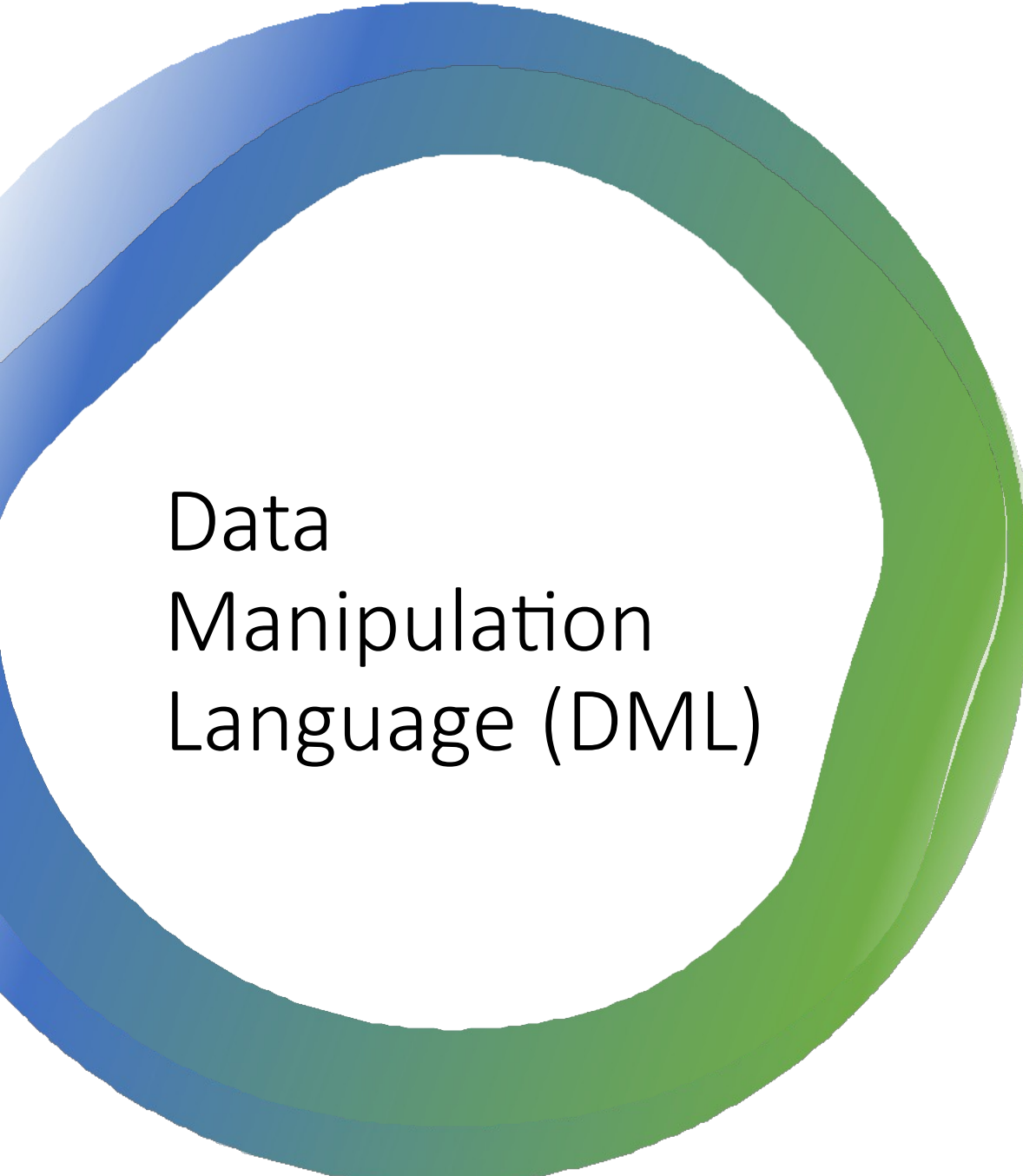
Rename:

- It is used to rename an object.

Comment:

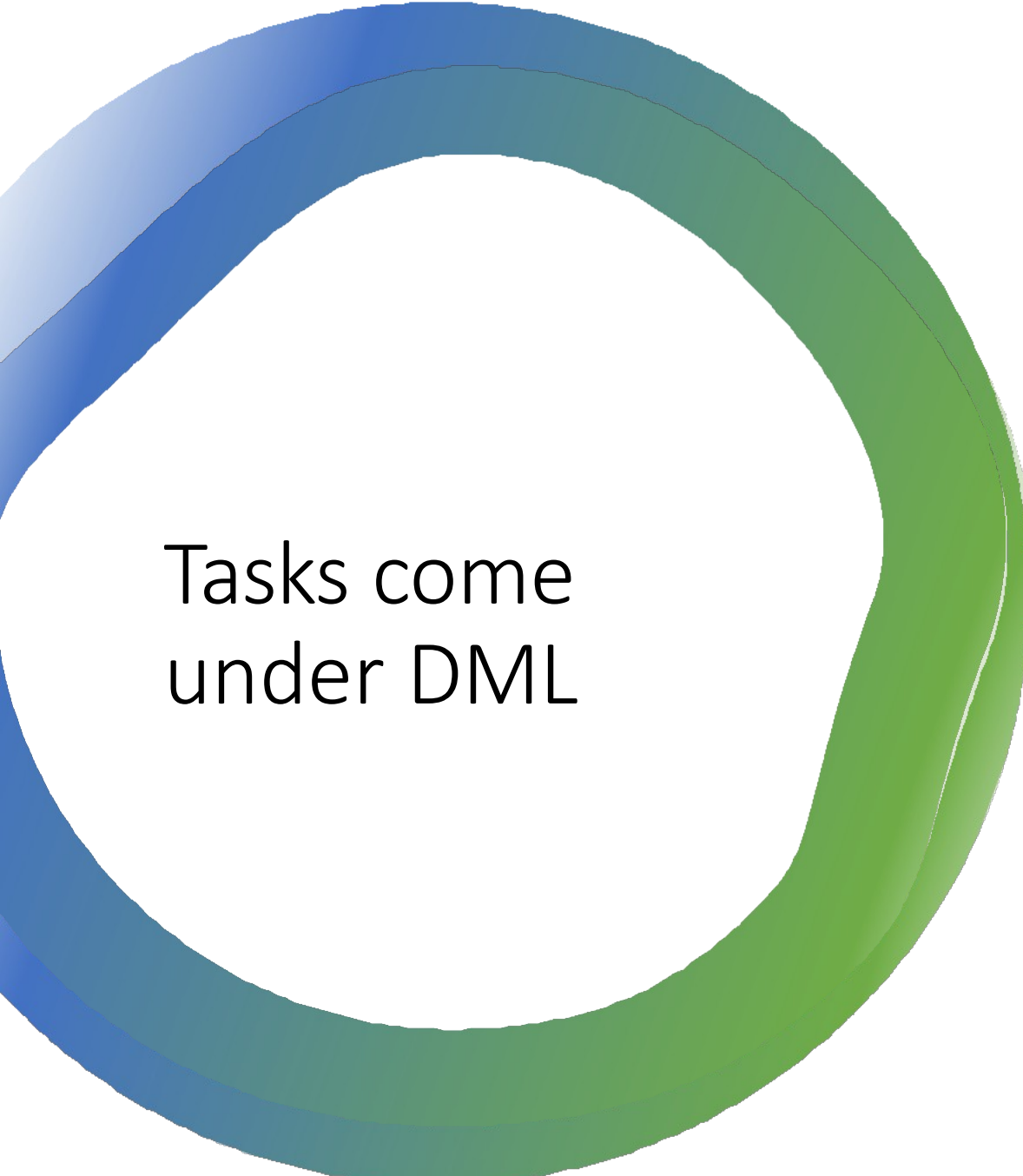
- It is used to comment on the data dictionary.

These commands are used to update the database schema that's why they come under DDL.



Data Manipulation Language (DML)

- **DML** stands for **Data Manipulation Language**.
- It is used for accessing and manipulating data in a database.
- It handles user requests.



Tasks come under DML

- It is used to retrieve data from a database.

Select:

- It is used to insert data into a table.

Insert:

- It is used to update existing data within a table.

Update:

- It is used to delete all records from a table.

Delete:

- It performs UPSERT operation, i.e., insert or update operations.

Merge:

- It is used to call a structured query language or a Java subprogram.

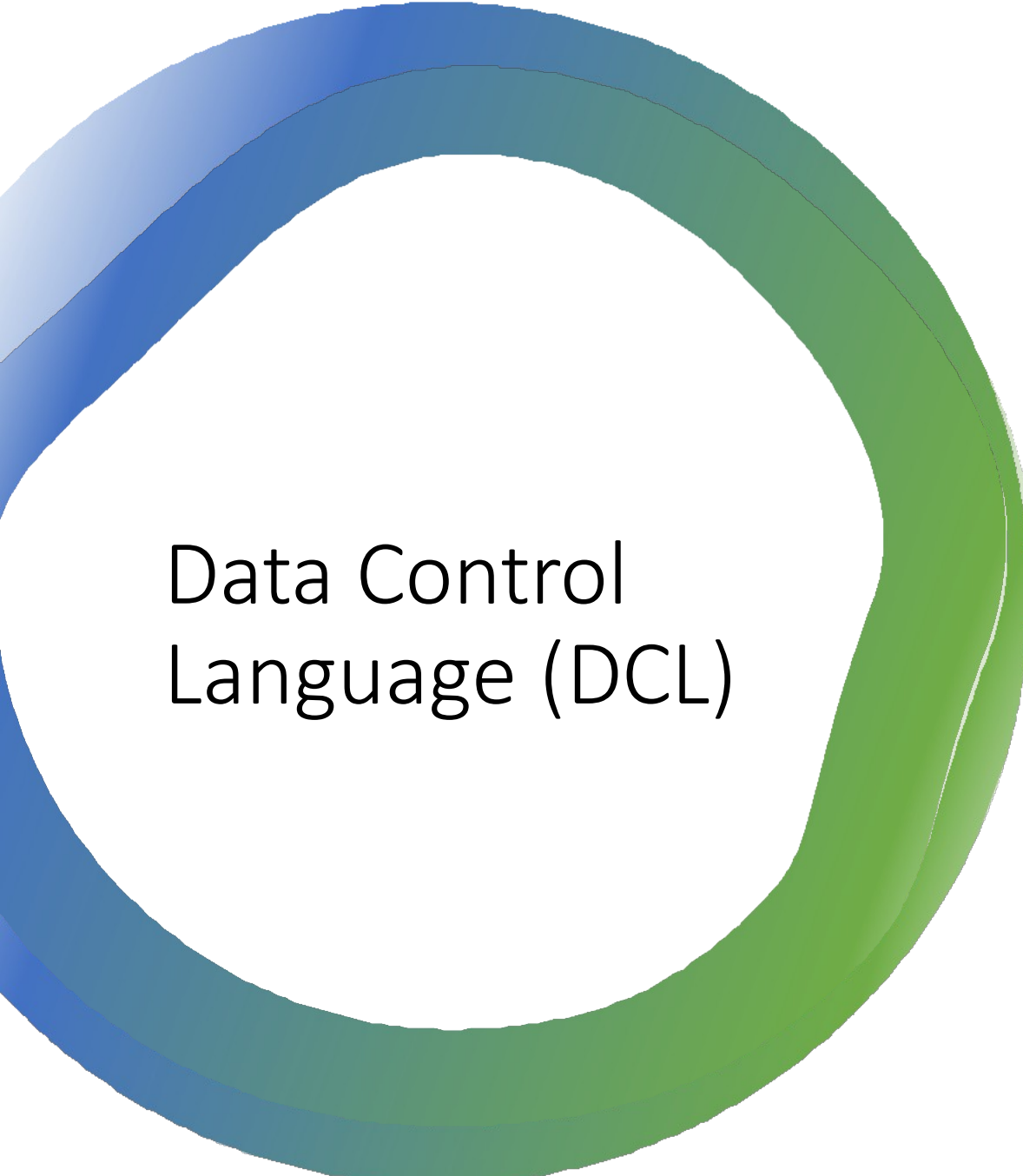
Call:

- It has the parameter of explaining data.

Explain Plan:

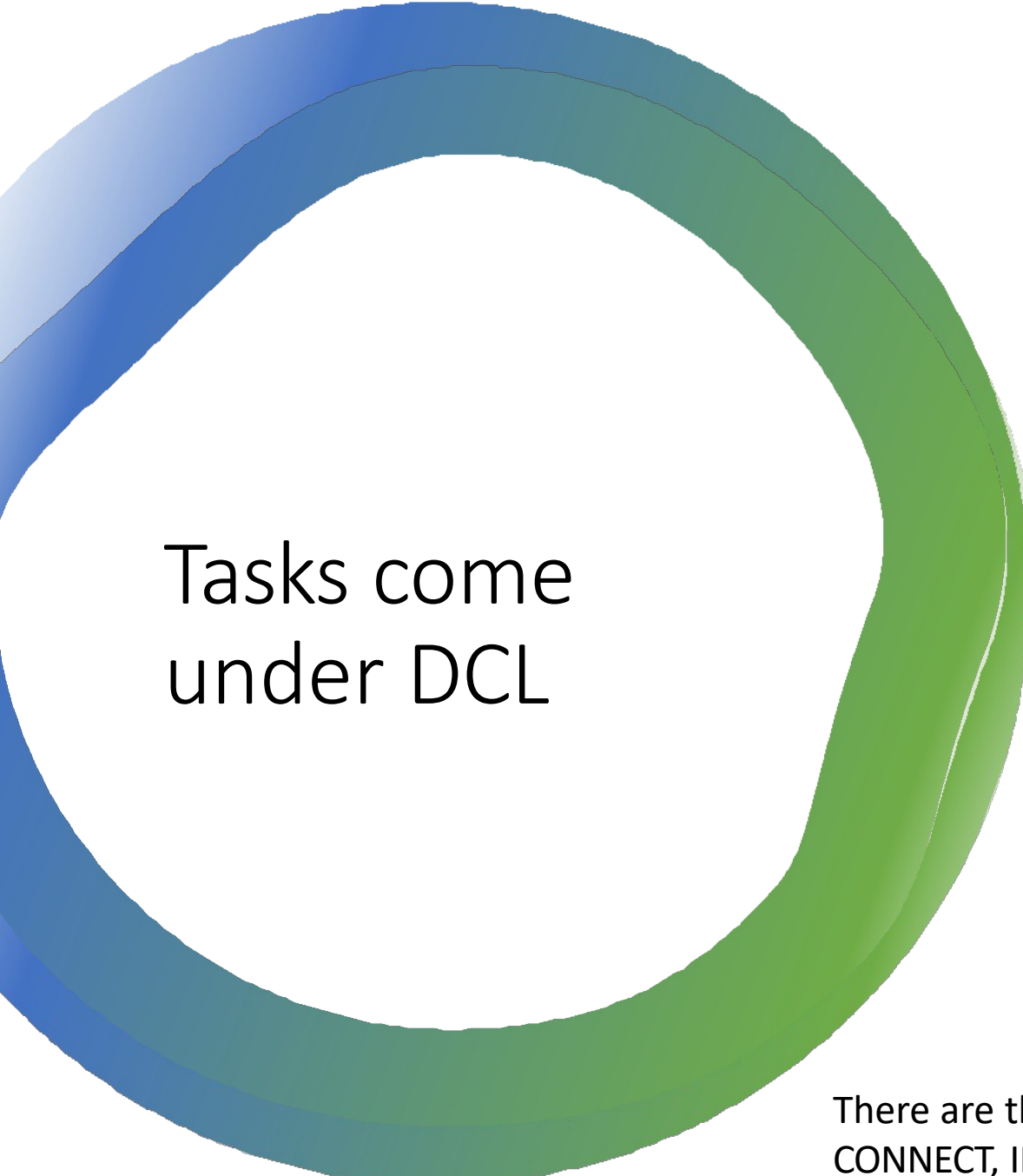
- It controls concurrency.

Lock Table:



Data Control Language (DCL)

- **DCL** stands for **Data Control Language**.
- It is used to retrieve the stored or saved data.
- The DCL execution is transactional.
- It also has rollback parameters.
- (But in Oracle database, the execution of data control language does not have the feature of rolling back.)



Tasks come
under DCL

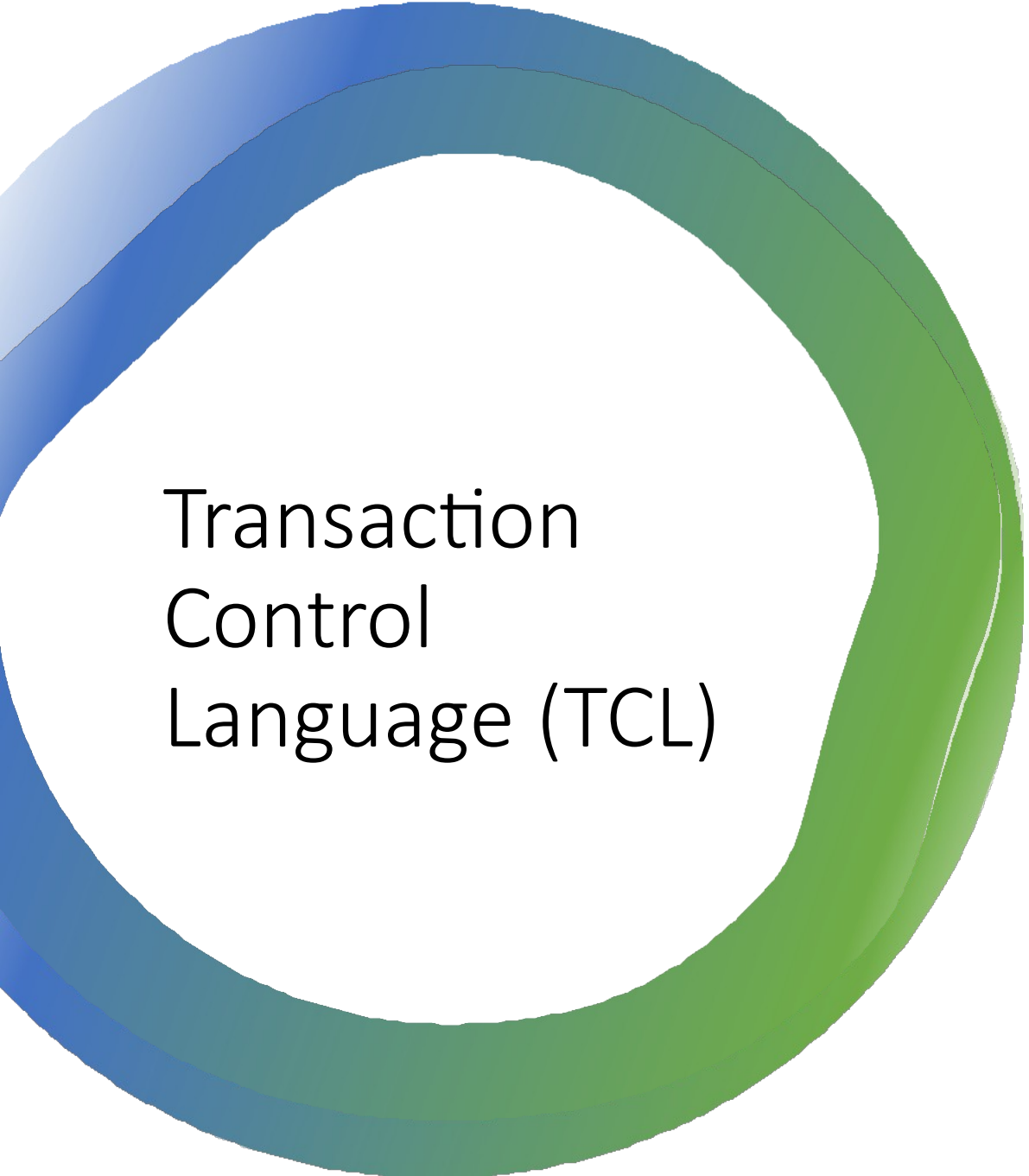
Grant:

- It is used to give user access privileges to a database.

Revoke:

- It is used to take back permissions from the user.

There are the following operations which have the authorization of Revoke: CONNECT, INSERT, USAGE, EXECUTE, DELETE, UPDATE and SELECT.



Transaction Control Language (TCL)

- TCL is used to run the changes made by the DML statement.
- TCL can be grouped into a logical transaction.



Tasks come
under TCL

